



OSI Reference Model

OSI Model

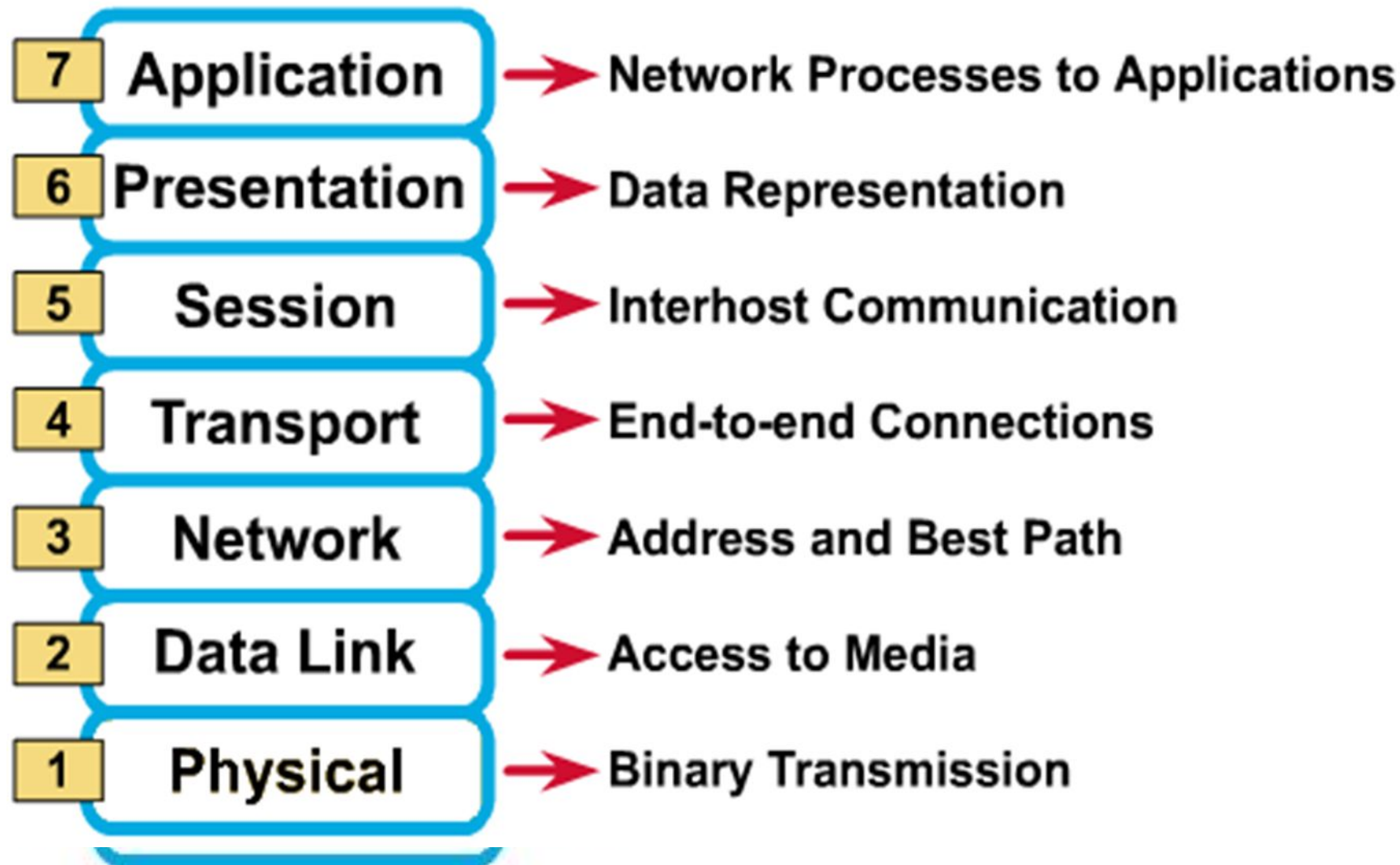


- The OSI model is a layered framework for the design of network systems that allows **communication between all types of computer systems**.
- It consists **of seven separate but related layers**.
- Each layer defines a part of the process of moving information across a network.
- An OSI model provides a solid basis for understanding data communications.

ISO –OSI Open Systems Interconnection Model

- OSI describes how information flow from **application programmes** -> through a network medium (such as **wire**)-> to **another application** located on another network.
- **moving information** between computers **SEVEN** smaller layering.
- helps to **develop standards** so that **all of our hardware and software talks** nicely to each other.
- **compatible with current and future technologies.**

7 Layers of OSI Model



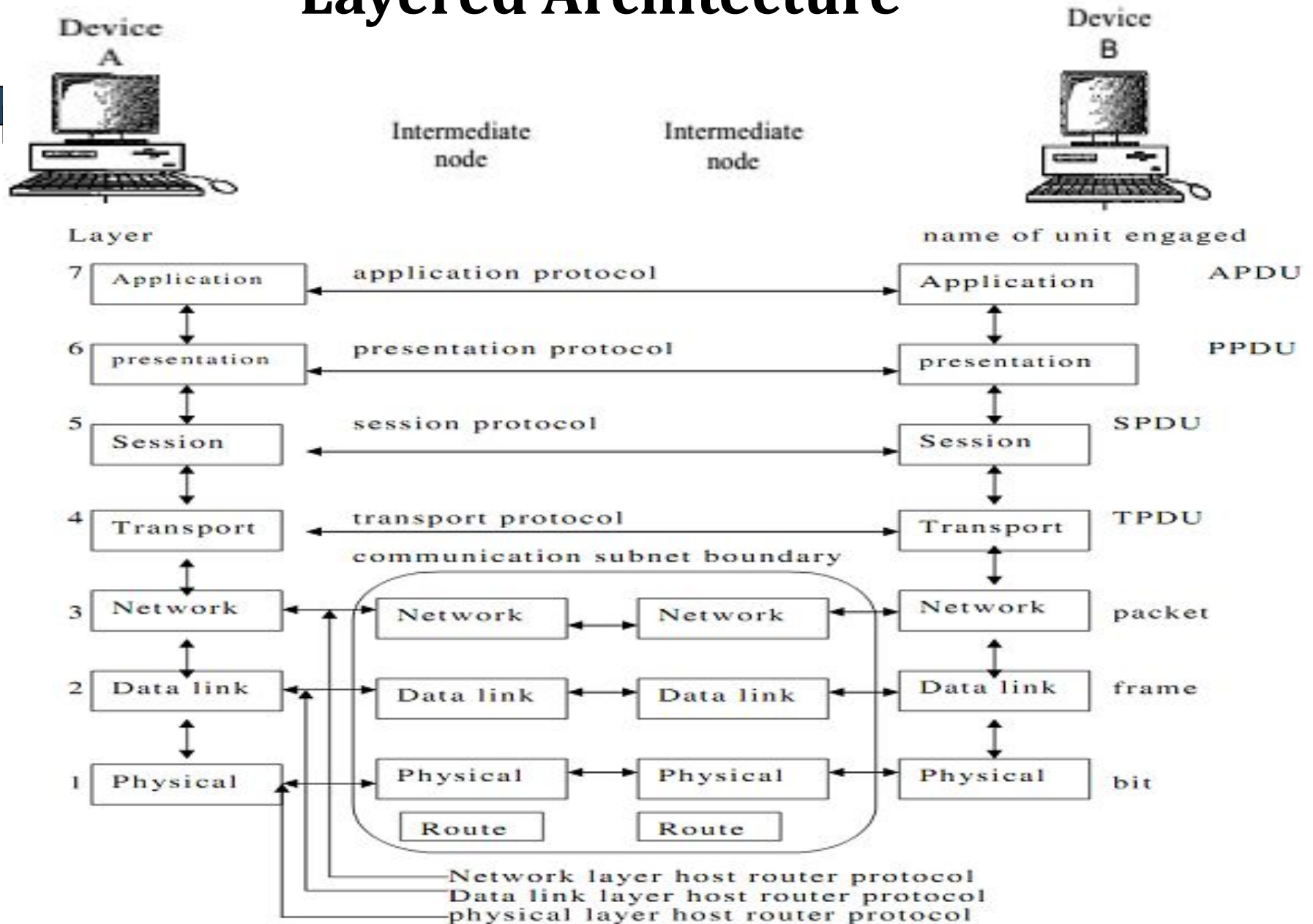
What is Layered Architecture?

- **1. LA reduce the design complexity**, most of the networks are organized as a series of layers or levels, each one build upon one below it.
- It **divide the design into small pieces**.
- **Each layer adds to the services provided by the lower layers** => highest layer is provided a full set of services to manage communications and run the applications. For example, Layer 3 uses the services provided by layer 2 and provides services for layer 4.
- **benefits: modularity** and clear interfaces, i.e. open architecture and comparability between the different provider's components.

Layered Architecture

- The layers involved when a message is sent from **device A to device B**.
- As the message travels from A to B, it may pass through many **intermediate nodes**.
- These intermediate nodes usually involve only the **first three layers** of the OSI model.
- Each layer performs **unique and specific task** and it **offers services to the layer above it**.
- For example, **Layer 3 uses the services provided by layer 2** and **provides services for layer 4**.

Layered Architecture



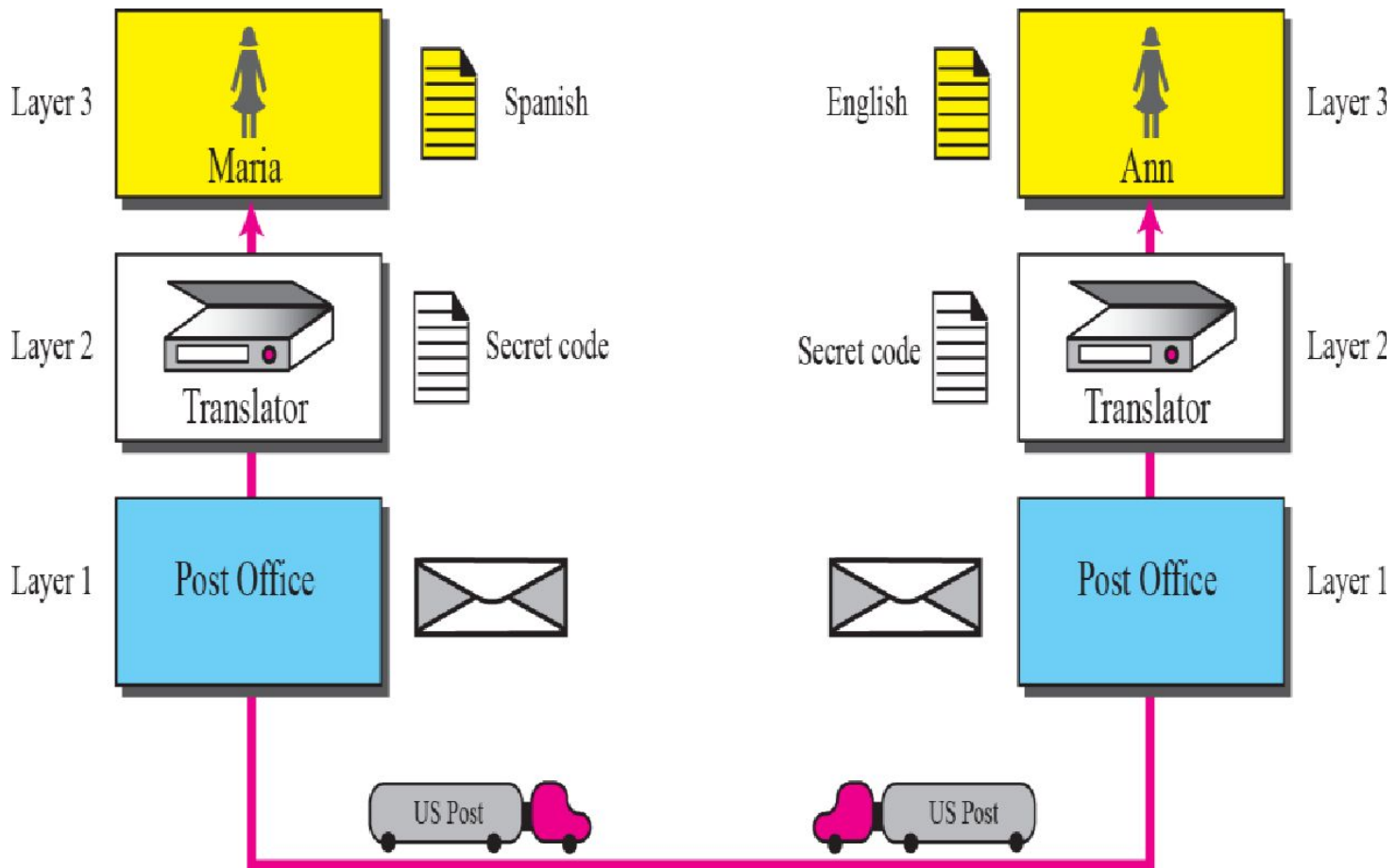


- Between machines, **layer x on one machine communicates with layer x on another machine.**
- This communication is governed by an agreed-upon series of rules and conventions called **protocols.**
-

REASONS Layered Architecture



- 1. functions well defined**
- 2. flexibility to modify**
- 3. lower layer offers certain services to its upper layer.**
- 4. considerable cost savings and managerial benefits.**
- 5. Addition of new services become easy.**
- 6. break complex problems into smaller**
- 7. helps development**



Peer-to-Peer processes

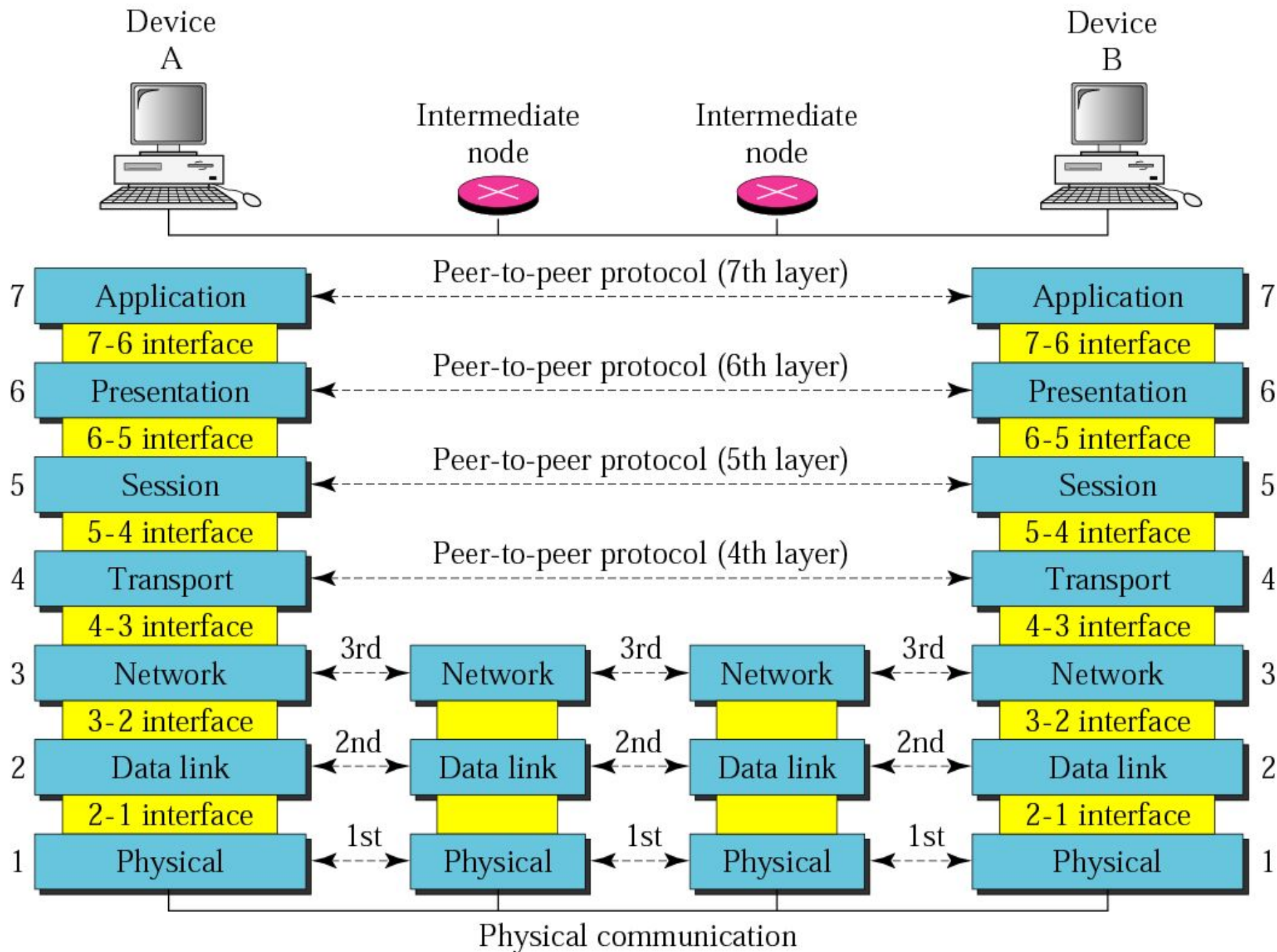
- The processes on each machine that communicate at a given layer are called **peer-to-peer processes**.
- At the **physical layer**, communication is direct:
Device A sends a stream of bits to device B
(through intermediate nodes).
- At higher layers each layer in the sending machines **adds its own information to the message it receives from the layer just above it** and **passes the whole package to the layer just below it**.

Peer-to-Peer processes

- The information added by each layer is in the form of **headers or trailers**.
- At layer 1 the entire package is converted to a form that can be transferred to the receiving machine.
- At the **receiving machine, the message is unwrapped layer by layer**, with each process receiving and removing the data meant for it.
- For example, layer 2 removes the data meant for it, then passes the rest to layer 3.
- Layer 3 then removes the data meant for it and passes the rest to layer 4, and so on.

Interfaces Between Layers

- An interface defines the **operations and services offered by lower layer** to the upper layer.
- This is an interface between each pair of adjacent layers.
- The passing of the data and network information down through the layers of the sending device and receiving device is made possible by an interface between each pair of adjacent layers.

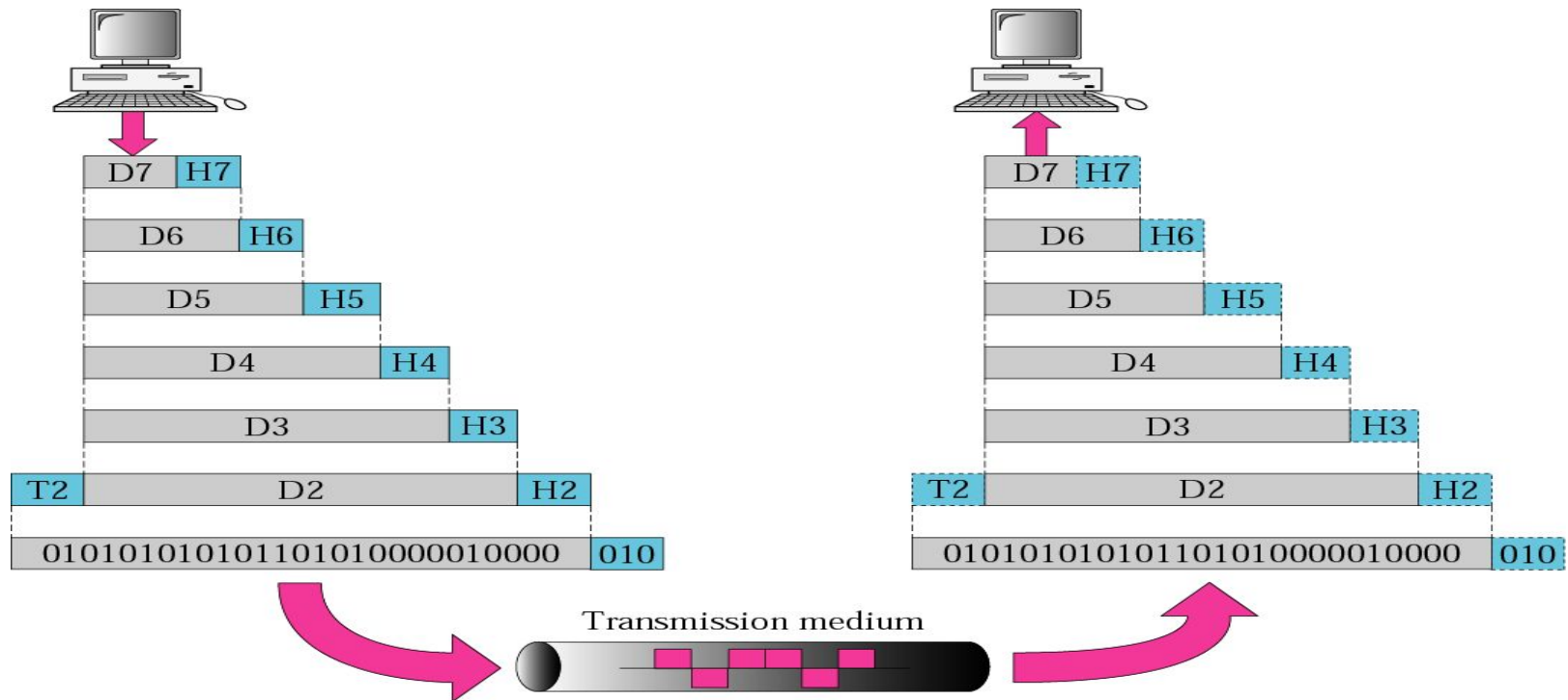


Data Encapsulation

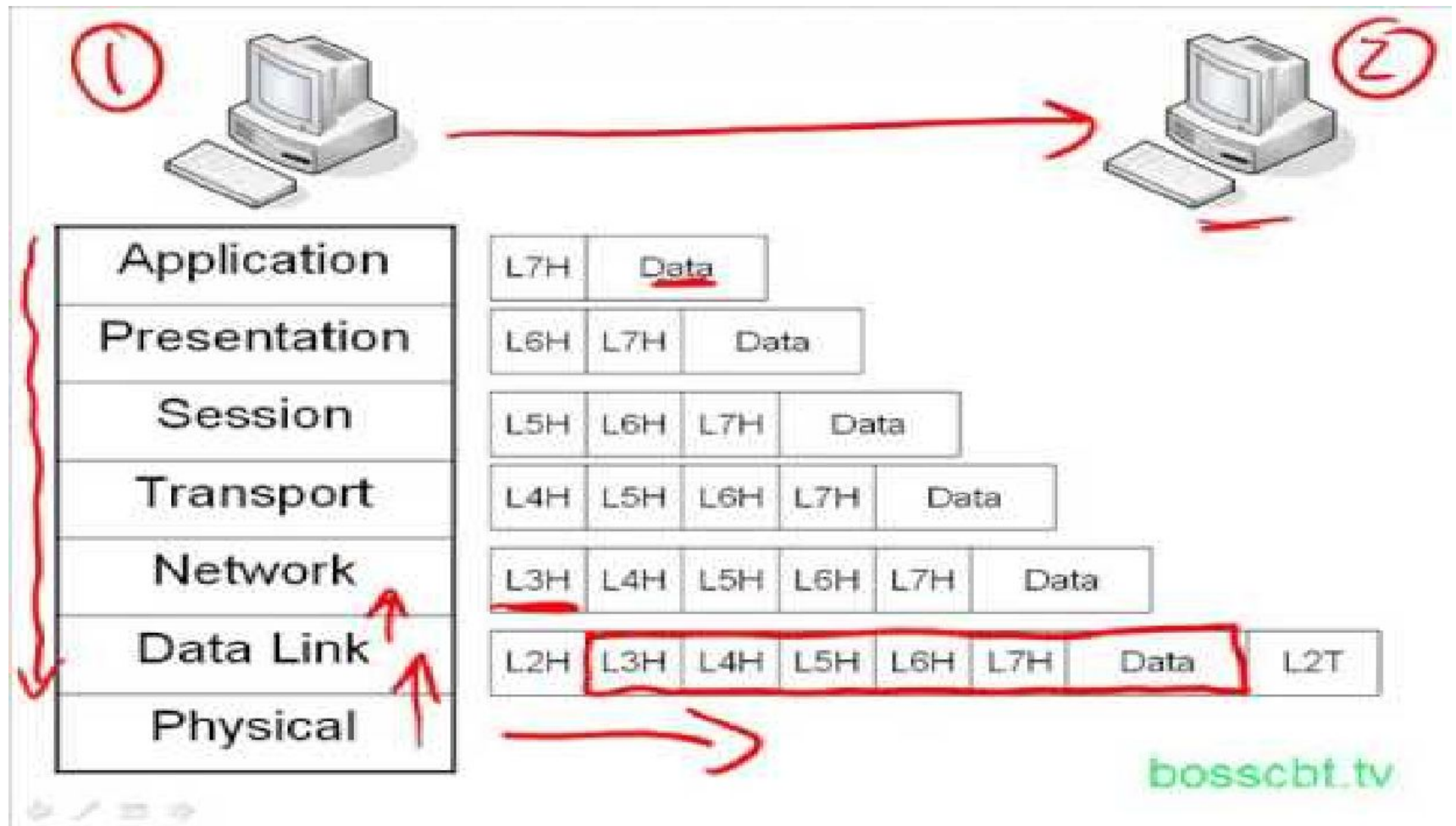
- The outgoing information will travel down through the top layers to the lowest layer.
- While moving down on the source machine, it acquires all the **control information** which is required to reach the destination machine.
- The control information is in the form of **Headers and Trailer** which surrounds the data received from the layer above.
- This process of adding headers and trailers to the data is called as **data encapsulation**.

Data Encapsulation

- The headers and trailers contain control information.
- The headers and trailers form **the envelope** which carries the message to the desired destination.



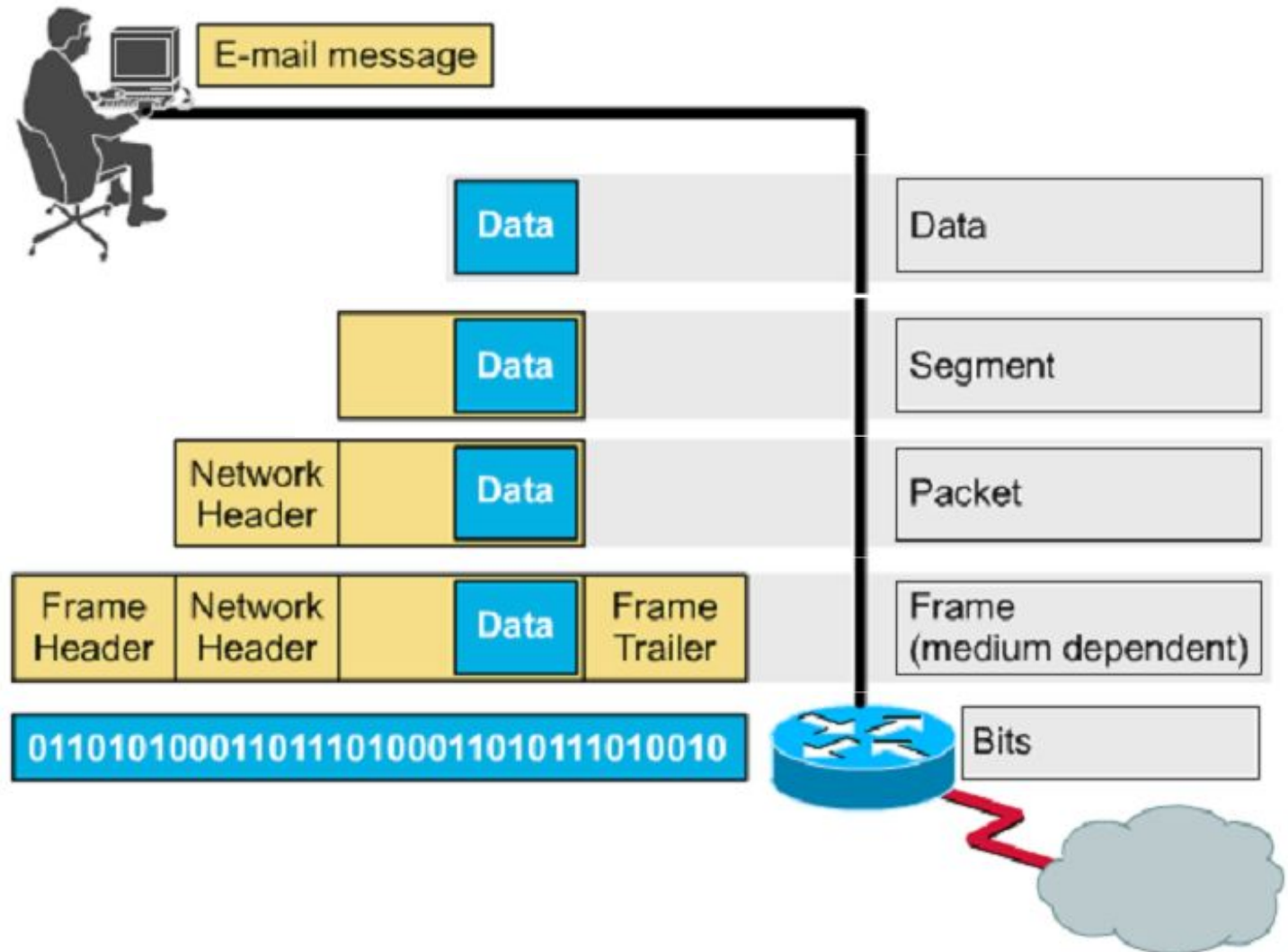
The protocols operating at the various layers work together to supply a unified quality of service. Each protocol layer provides a service to the layers directly above and below it. The **process of adding the headers and trailers to the data** is called as **data encapsulation**.



Data Encapsulation

- D7 means the data unit at layer 7, D6 means the data unit at layer 6, and so on.
- The process starts at layer 7 (the application layer), then moves from layer to layer in descending, sequential order.
- At each layer, a **header**, or possibly a **trailer**, can be added to the data unit.
- Commonly, the trailer is added only at layer 2.
- When the formatted data unit passes through the physical layer (layer 1), it is changed into an electromagnetic signal and transported along a physical link.

Encapsulation example: E-mail



Draw OSI reference model.

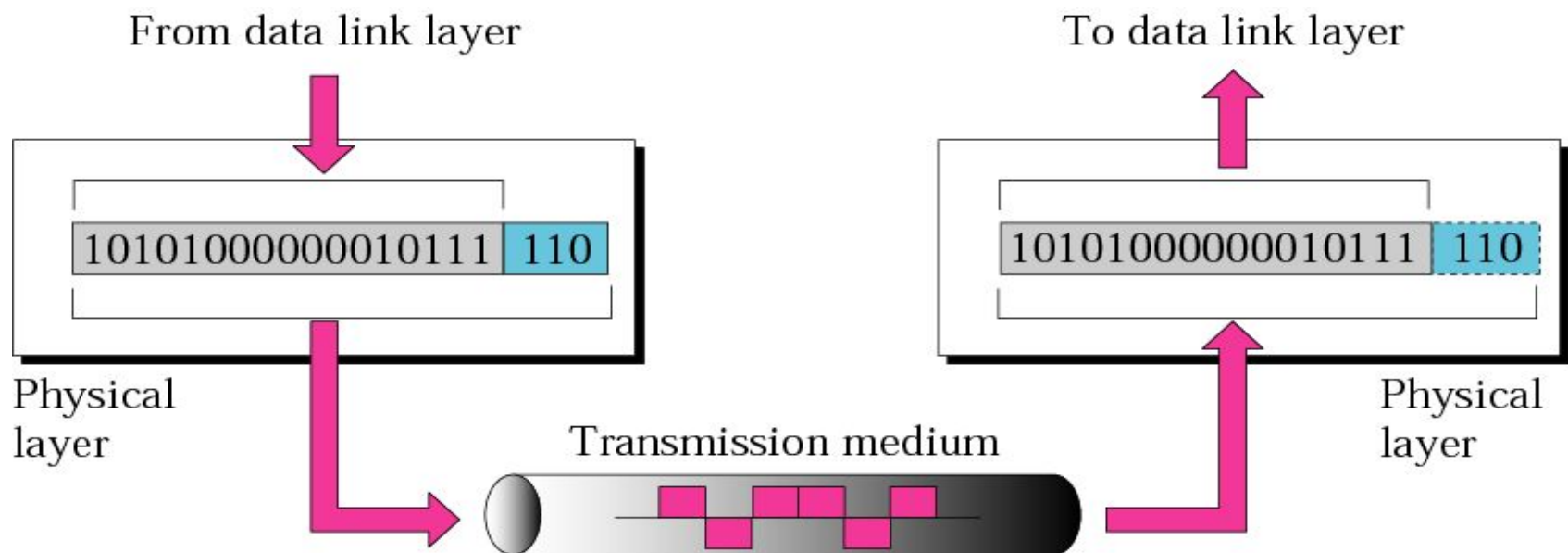
Explain working

- It provides way to understand **how internet work** operates.
- It gives **guideline for creating network standard**.
- OSI model has following 7 layers as
 - 1 Physical layer,
 - 2 data link layer,
 - 3 Network layer,
 - 4 Transport layer,
 - 5 Session layer,
 - 6 Presentation layer,
 - 7 Application layer.

1. Physical Layer

- It co-ordinates the functions required to transmit bit stream over physical medium.
- Provides physical interface for transmission of information.
- Defines rules by which bits are passed from one system to another on a physical communication medium.
- Covers all - **mechanical, electrical, functional and procedural** - aspects for physical communication.

1. Physical Layer



The physical layer is responsible for the movement of individual bits from one hop (node) to the next.

Functions

- **Physical characteristics** of interfaces and medium.
 - It also defines the type of transmission medium.
- **Representation of bits.**
 - sequence of 0s or 1s
- **Data rate.**
- **Synchronization of bits.**
 - Sender and Receiver must be synchronized
- **Physical topology**
 - Mesh, Ring, Star, etc.
- **Transmission mode.**
 - Simplex, Half duplex, Duplex

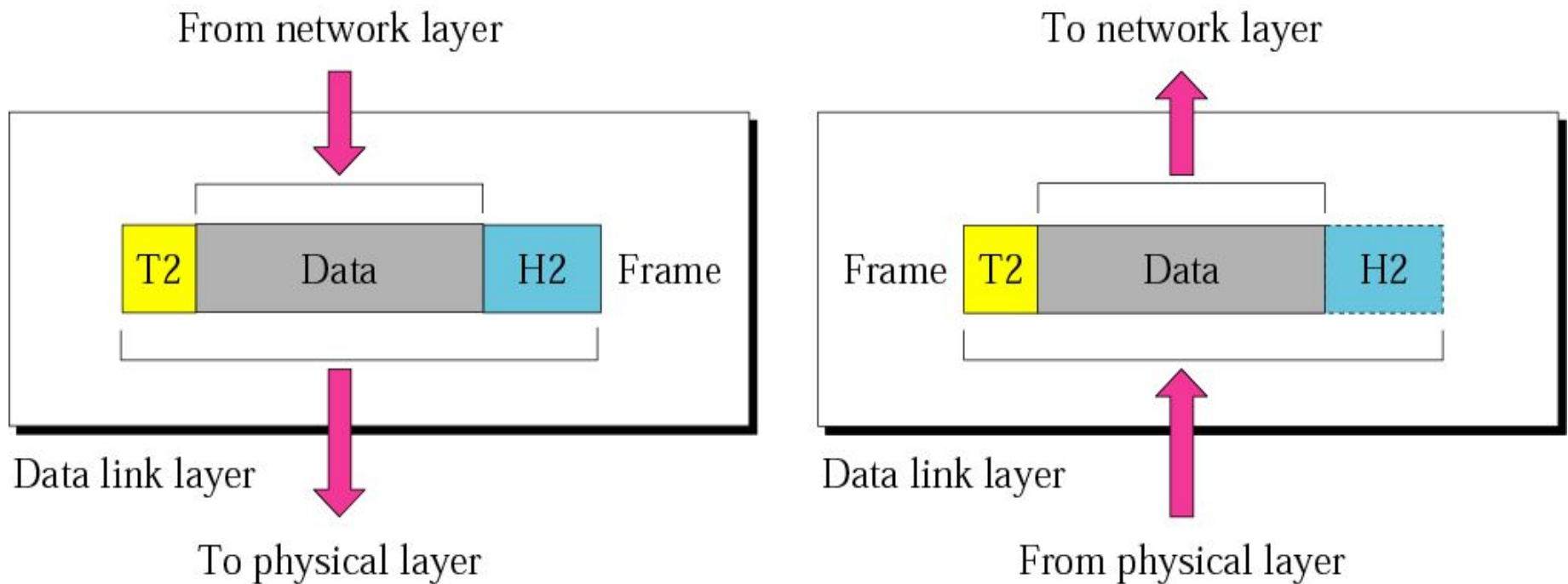
1. Physical Layer

- What are the Physical Layer components on my computer?
- **NIC**
 - Network Interface Card
 - Has a unique 12 character Hexadecimal number permanently burned into it at the manufacturer.
 - The number is the MAC Address/Physical address of a computer
- **Cabling**
 - Twister Pair
 - Fiber Optic
 - Coax Cable

2. Data Link Layer

- Data link layer attempts to **provide reliable communication** over the physical layer interface.
- **Breaks the outgoing data into frames** and re-assemble the received frames.
- Create and detect frame boundaries.
- **Handle errors** by implementing an acknowledgement and retransmission scheme.
- **Implement flow control.**

2. Data Link Layer



*The data link layer is responsible for moving **frames** from one hop (node) to the next.*

Data Link Layer Functions

- **Framing-**

- divides the stream of bits into manageable data units called frames.

- **Physical addressing-**

- adds a header to the frame to define the sender and/or receiver of the frame.

- **Flow control-**

- imposes a flow control mechanism to avoid overwhelming the receiver.

- **Error control-**

- adds mechanisms to detect and retransmit damaged or lost frames.

- **Access control-**

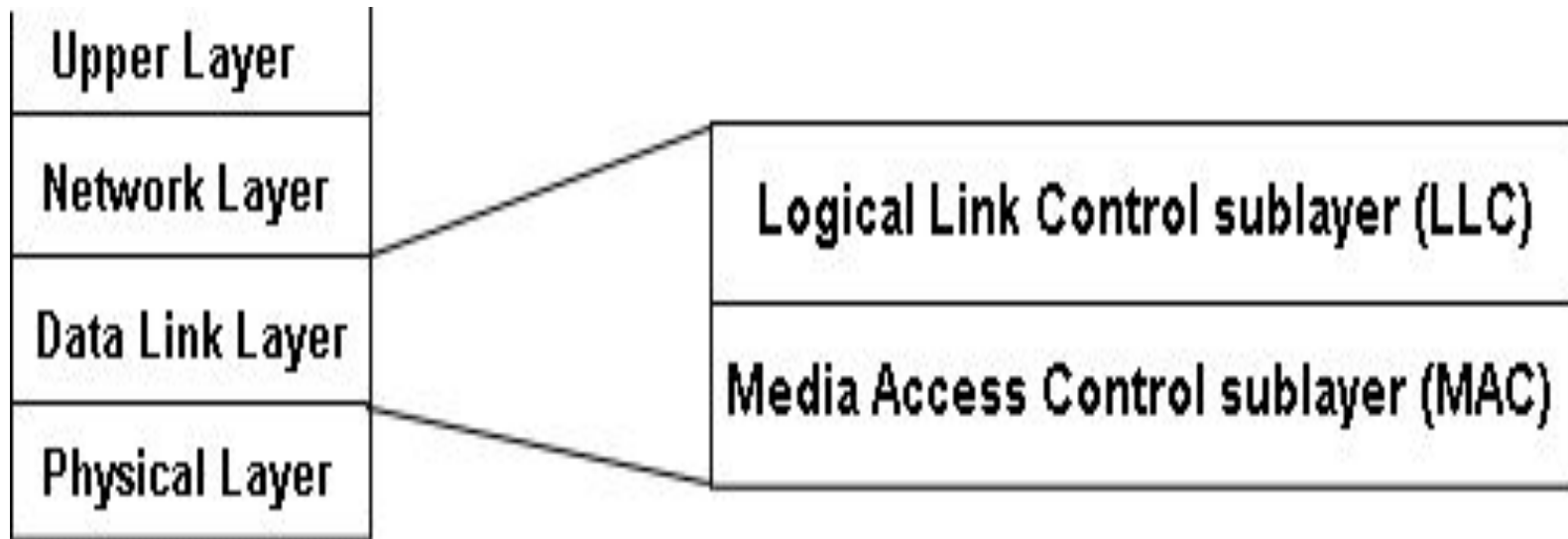
- determine which device has control over the link at any given time.

Data Link Layer Functions

- **Link establishment and termination:**
 - establishes and terminates the logical link between two nodes.
- **Frame sequencing:**
 - transmits/receives frames sequentially.
- **Frame acknowledgment:**
 - provides/expects frame acknowledgments.

Data Link Layer –Sub Layers

- DLL is divided into two Sub-Layers
 - **LLC Sub Layer**
 - **MAC Sub Layer**



Logical Link Control Sub Layer

- It is upper portion of the Data Link layer.
- Performs **Flow control** and **management of connection errors**.
- LLC supports three types of connections:

1. Unacknowledged connectionless service:

- does not perform reliability checks or maintain a connection, very fast, most commonly used

2. Connection oriented service:

- once the connection is established, blocks of data can be transferred between nodes until one of the node terminates the connection.

3. Acknowledged connectionless service:

- provides a mechanism through which individual frames can be acknowledged.

Media Access Control Sub Layer

- This sub layer contains methods to **regulate the timing** of data signals and **eliminate collisions**.
- The MAC sub layer determines where one frame of data ends and the next one starts - **frame synchronization**.
- There are four means of frame synchronization:
 1. Time based,
 2. Character counting,
 3. Byte stuffing and
 4. Bit stuffing.

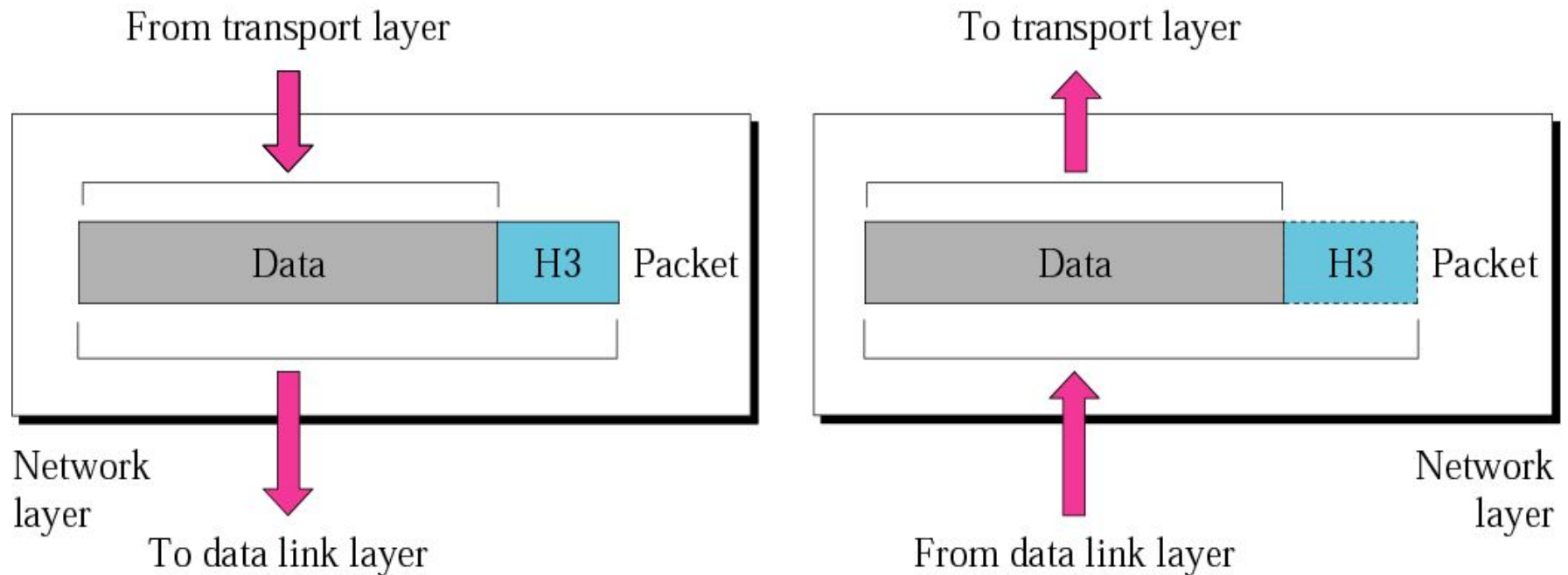
3. Network Layer

- It is responsible for source to destination delivery of individual **packets** across multiple networks.
- Defines the **most optimum path** the packet should take from the source to the destination
- Defines **logical addressing** so that any endpoint can be identified.

3. Network Layer

- **Handles congestion** in the network.
- Facilitates interconnection between heterogeneous networks (Internetworking).
- The network layer also defines how to fragment a packet into smaller packets to accommodate different media.

3. Network Layer



*The network layer is responsible for the delivery of individual **packets** from the source host to the destination host.*

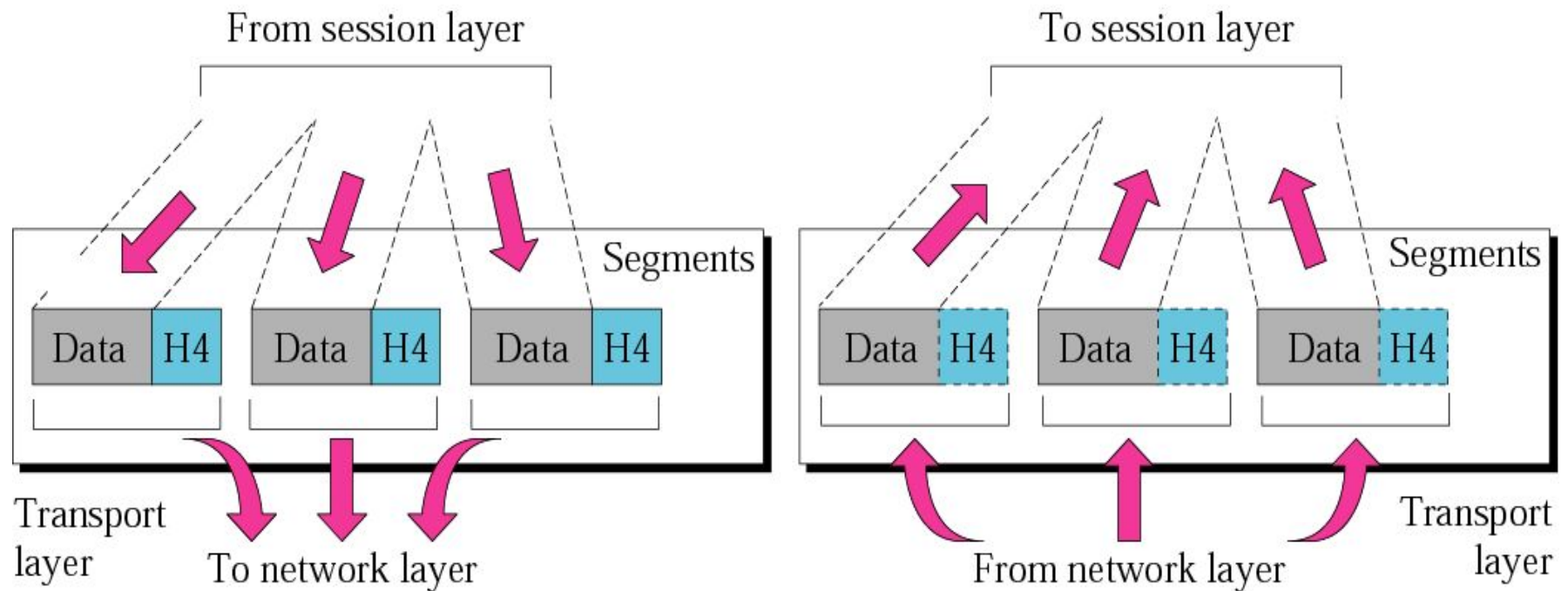
Functions of Network Layer

- 1. Logical addressing**
- 2. Routing.**
- 3. Congestion control**
- 4. Accounting and billing**
- 5. Address transformation**
- 6. Source host to destination host error free delivery of packet.**

4. Transport Layer

- Purpose of this layer is to provide a **reliable mechanism** for the **exchange of data between two processes** in different computers.
- Ensures that the data units are **delivered error free**.
- Ensures that data units are **delivered in sequence**.
- Ensures that there is **no loss or duplication** of data units.
- Provides **connectionless or connection oriented** service.

4. Transport Layer



*The transport layer is responsible for the delivery of a **message** from one process to another.*

Functions Transport Layer

1. Service point addressing
2. Segmentation and reassembly
3. Connection control
4. **Flow control**: Flow control is performed end to end
5. Error control

5. Session Layer

- Session layer provides mechanism for **controlling the dialogue** between the two end systems.
- It defines how to **start, control and end** conversations (called sessions) between applications.
- This layer requests for a logical connection to be established on an end-user's request.
- Any necessary **log-on or password validation** is also handled by this layer.

Functions of Session Layer

- 1. Dialog control**
- 2. Synchronization, session and sub session**
- 3. Session closure**

6. Presentation Layer

- Presentation layer defines the **format** in which the data is to be exchanged between the two communicating entities.
- Also handles data compression and data encryption (cryptography).

6. Presentation Layer

- **Functions of Presentation layer:**
- **Translation:** presentation layer is responsible for converting various formats into required format of the recipient.
- **Encryption:** Data encryption and decryption is done by presentation layer for security.
- **Compression and Decompression:** data to be transform compressed while sending and decompress while receiving for reducing time of transmission.

7. Application Layer

- Application layer interacts with application programs and is the highest level of OSI model.
- Application layer contains management functions to support distributed applications.
- Examples of application layer are applications such as
 - file transfer,
 - electronic mail,
 - remote login etc.

7. Application Layer

- **Functions of Application layer:**

1. Network virtual terminal
2. File transfer access and management
3. Mail services and directory services

Horizontal Communication

- The horizontal communication is the **Logical connection** between the layers.
- There is **no direct communication** between them.
- Information included in each protocol header by the **transmitting system** is a message that will be carried to the same protocol in the **destination system**.
- For two computers to communicate over a n/w, the **protocol used at each layer** of the OSI model in the transmitting system **must be duplicated** at the receiving system.

Horizontal Communication

- When the packet reaches at its destination, the process by which the headers are applied at the source is reversed in receiver to get information.

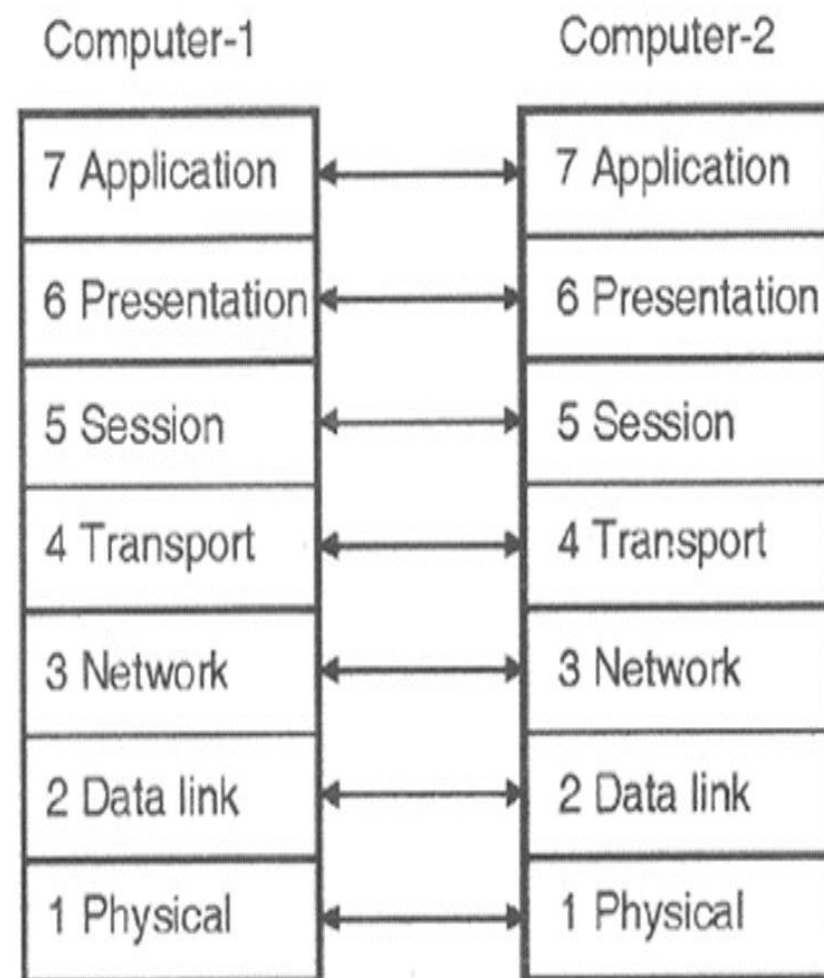


Fig: Horizontal Communication in OSI Model.

Vertical Communication

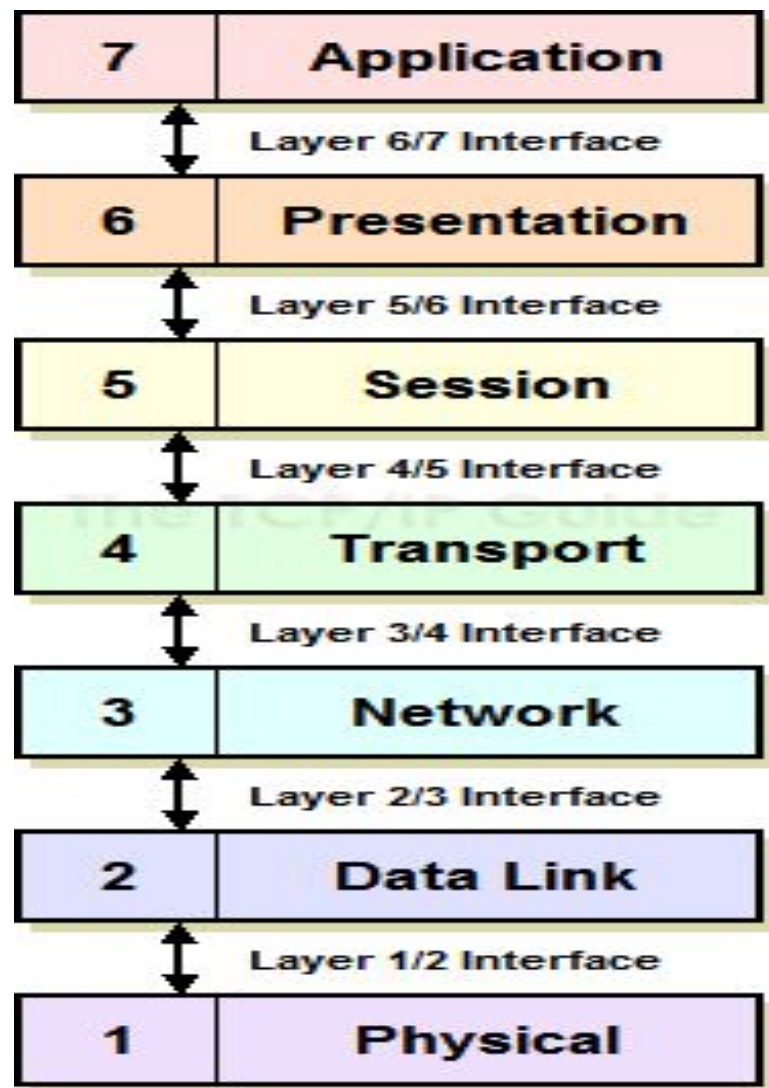
- The header information also enables **each layer to communicate** with the layer above & below it.

Eg. The n/w layer will communicate with the **data link layer & transport layer**.

- This **interlayer communication** is called vertical communication.
- When a system receives a packet & passes it up through various layers the data link layer protocol header includes a field which specifies the name of n/w layer protocol to be used to process the packet.

Vertical Communication

- The n/w layer protocol header will specify the name of transport layer protocol to be used to process the packet.



Connection Oriented Services

- Connection-oriented communication includes the steps of **setting up a call** from one computer to another, **transmitting/receiving data**, and then **releasing the call**, just like a voice phone call.
- However, the network connecting the computers is a packet switched network.
- Connection-oriented communication is done in one of two ways over a packet switched network: **with and without virtual circuits**.
- We can **resend** the message in connection oriented service if there is an error at the receivers end.
- Example of connection oriented is **TCP (Transmission Control Protocol)** protocol.

Connectionless Services

- **Connectionless** communication is just packet switching where **no call establishment** and **release occur**.
- A message is broken into packets, and each packet is transferred separately.
- Moreover, the **packets can travel different route** to the destination since there is no connection.
- Connectionless service is typically provided by the **UDP (User Datagram Protocol)**.
- The packets transferred using UDP are also called **datagrams**.



Protocols

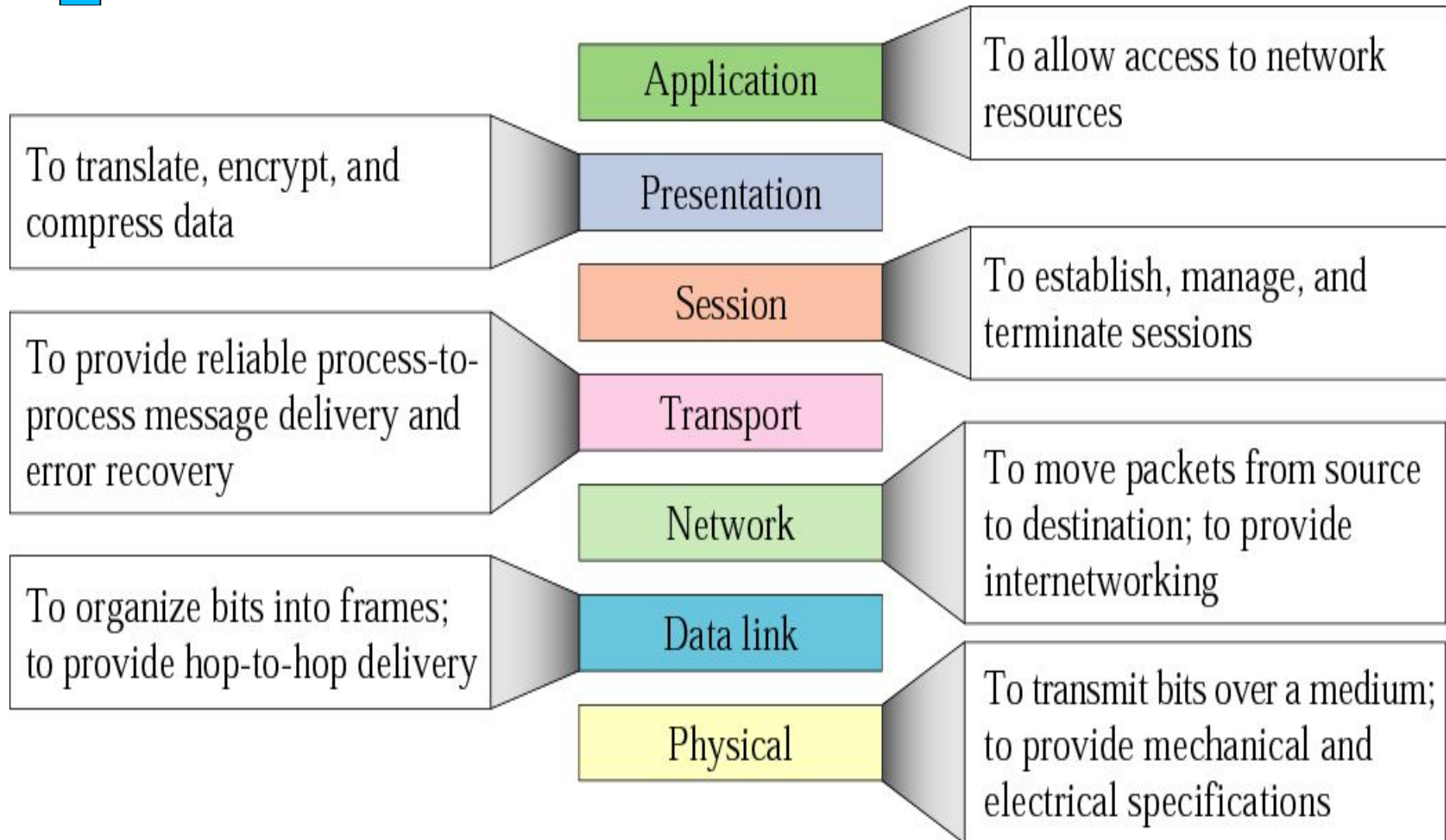
Protocols - Layer 7	HTTP, FTP, SMTP, POP, DNS, Telnet
Protocols - Layer 6	Graphics: TIFF, JPEG, GIF Text: ASCII, EBCDIC, Unicode, Encrypted Audio: Midi, MPEG, WAV, MP3 Video: Quicktime, AVI
Protocols - Layer 5	NFS - Network File System, RPC - Remote Procedure Call, ASP - Appletalk Session Protocol
Protocols - Layer 4	Connection Oriented (TCP), Connectionless (UDP)
Protocols - Layer 3	Routing (IP)
Protocols - Layer 2	MAC - Media Access Control, LLC - Logical Link Control
Protocols - Layer 1	Organizations: IEE, TIA/ETA, ANSI, etc. Cable (ie. RJ45)



Devices

Devices - Layer 7	Hosts, PC, Servers, Mobile Phones
Devices - Layer 3	Routers
Devices - Layer 2	Bridges, Switches, NIC (Layers 1 & 2)
Devices - Layer 1	Hubs, NIC (Layers 1 & 2) Media: Coax, Fiber, Twisted Pair, Wireless

Summery



Remembering the 7 Layers

7 - Application	All	Away	
6 - Presentation		People	Pizza
5 - Session		Seem	Sausage
4 - Transport	To	Throw	
3 - Network		Need	Not
2 - Data Link	Data	Do	
1 - Physical		Processing	Please

Que.: Name the layer which is associated with the transmission media.

Que: Write the name of layers that perform the following functions in OSI :

- I. Data Encryption**
- II. Error Correction**
- III. File Transfer**
- IV. Data Encoding**